

Nicholas Martis

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Faculty of Mathematics and Physics, University of Ljubljana

RESEARCH INTERESTS

Observational galaxy formation and evolution; dust-obscured galaxies; galaxy cluster formation

RESEARCH POSITIONS

- **Assistant with a Doctorate** 2023 - Present
University of Ljubljana Ljubljana, Slovenia
 - CANUCS collaboration imaging working group lead - data release 
 - MINERVA collaboration MIRI imaging reduction lead
 - VENUS collaboration team member
- **Post Doctoral Fellow, Astronomy and Physics** 2020 - 2023
St. Mary's University Halifax NS, Canada
 - CANUCS collaboration imaging working group lead
- **Research Assistant** 2015 - 2020
Tufts University Medford MA, United States
 - 2 nights of near-infrared imaging with MOSFIRE at Keck telescope (Hawaii, USA)

EDUCATION

- **Tufts University** 2020
Ph.D. Astrophysics Medford MA, United States
 - Advisor: Danilo Marchesini
 - Thesis: "The Prevalence and Properties of Dusty Galaxies Through Cosmic Time"
- **Tufts University** 2016
M.S. Physics Medford MA, United States
 - Completed Graduate Professional Development Workshop Series
- **Boston College** 2014
B.S. Physics summa cum laude Chestnut Hill MA, United States
 - Second major: Philosophy

RESEARCH MENTORSHIP

Tijana Leban, master's thesis co-advisor, intracluster light in CANUCS clusters	2025-Present
Oskar Mlakar, master's student, distribution of globular clusters in Bullet Cluster	2025-Present
Miha Brvar, undergraduate student, intracluster light in Bullet Cluster	2025
Tess Grindlay, CSA co-op (undergraduate) student, emission line maps of cluster galaxies	2022

TEACHING

University Teaching for Graduate Students	2023
• Completed course on pedagogical technique	
Graduate Writing Consultant	2019-2020
• Provide writing tutoring to undergraduate and graduate students of all disciplines	
Mathematics and Physics Tutor	2018-2021
• Tutor high school and college students in pre-calculus, calculus, and introductory physics	
Teaching Assistant	2014-2015
• Lab and Recitation instructor for introductory Physics	

AWARDS

University of Ljubljana Outstanding Research	2025
• Project selected as one of the ten greatest research achievements of the University of Ljubljana 	
John F. Burlingame Graduate Fellowship in Physics	2019-2020
Graduate Student Travel Fund Award	2016, 2019

SERVICE

Journal Referee: Astronomy & Astrophysics, Astrophysical Journal

JWST External Panelist: Cycle 4, 5

Public data releases

- MAST High Level Science Products: CANUCS 🌐
- MINERVA Data Release 1 🌐
- Materials for individual CANUCS papers 🌐

OUTREACH

Golovec Evening at the Observatory – public lecture 🌐	2025
Kozmični Glasovi – collaboration between astronomers and Academy of Music at University of Ljubljana 🌐	2025
Salish Middle School Career Day	2024
CANUCS mini documentary. 🌐	2022

OBSERVATIONAL PROGRAMS

James Webb Space Telescope

1. MINERVA: Unlocking the Hidden Gems of the Distant Universe and Completing HST and JWST's Imaging Legacy with Medium Bands, 260 hours. PIs: Muzzin, Suess, Marchesini
2. A Unique Opportunity to Probe Lensing-Magnified Star Clusters in a Low-Mass Galaxy at $z_{\text{spec}} = 8.3$ with JWST, 50 hours. PI: Mowla
3. Vast Exploration for Nascent Unexplored Sources (VENUS), 252 hours. PI: Fujimoto
4. JUMPS: The JWST Ultimate Medium-band Photometric Survey, 42 hours. PI: Withers
5. Resolving galaxy building blocks at high- z : the comprehensive picture of internal physical properties in an ultra-low-mass major merger system at $z=5.2$, 24 hours. PI: Asada
6. JWST in Technicolor: Finding and Mapping the Most Extreme Star Forming Galaxies in the Epoch of Reionization with Medium and Narrow Bands, 44 hours. PI: Muzzin
7. Silver Bullet for Dark Matter, 17 hours. PI: Rhitaršič, Bradač, Sawicki
8. Completing the SMACS 0723 NIRISS WFSS EROs, 2 hours. PI: Noirot
9. Distant Sparkles: Are We Seeing Ancient Globular Clusters at Cosmic Noon?, 25 hours. PIs: Mowla, Iyer

Hubble Space Telescope

1. The Final Frontier: HST and JWST Exploration of Galaxies Across Cosmic Epochs, 56 orbits. PI: Bradač

CONFERENCES/WORKSHOPS

Unraveling Little Red Dots: Linking JWST Discoveries with Simulations to Understand Early Galaxies, Trieste, Italy	October 2025
Kaba Kada: Exploring the First Billion Years of the Universe, Port Douglas, Australia	September 2025
Peering Through an Opaque ISM, Ferrara, Italy	September 2025
Quo Vadis Galaxy Evolution? Heidelberg, Germany	July 2025
From the First Stars to the Epoch of Reionization, Trieste, Italy	April 2025
First Galaxies: Building Blocks of Galaxies Across Cosmic Time, Oxford, England	April 2025
Unveiling the physics of early galaxy and black hole formation with JWST, Paris, France	December 2024
European Astronomical Society Annual Meeting, Padova, Italy	July 2024
Extreme galaxies in their extreme environments at extremely early epochs, Reykjavik, Iceland	April 2024
First Year of JWST Science Conference, Space Telescope Science Institute, MD USA	September 2023
First Science Results from JWST, Virtual	December 2022
CASCA 2022: Canadian Astronomy in the Roaring 2020's, Virtual	May 2022
Large-Volume Spectroscopic Analyses of AGN and Star Forming Galaxies in the Era of JWST, Virtual	March 2022
SAZERAC SIPS: Early Galaxy Formation Near and Far, Virtual	November 2021
CASCA 2021 AGM, Virtual	May 2021
The Art of Measuring Galaxy Physical Properties, Milano, Italy	November 2019
Graduate Research Excellence at Tufts (GREAT), Tufts University, MA USA	June 2018
User Training in JWST Data Analysis, Space Telescope Science Institute, MD USA	November 2016
Deconstructing Galaxies at Cosmic Noon: The Present and Future of Deep Spectroscopic Surveys at High Redshift, Lorentz Center, Netherlands	August 2016

SEMINARS/TALKS

FCAD Astronomy Seminar, UMass Amherst, USA	December 2025
Astrodebata, University of Ljubljana, Slovenia	February 2023
Canadian Virtual Astronomy Seminar	November 2022
Astronomy and Physics Colloquium, St. Mary's University, Canada	February 2021
Seminar, Brera Astronomical Observatory, Milano, Italy	November 2019
Astronomy Seminar, University of California Riverside, CA USA	April 2019
Lunch Talk, Carnegie Observatories, CA USA	April 2019
Lunch Talk, University of California Los Angeles, CA USA	April 2019

SELECTED PRESS

1. $z \sim 8.6$ "little red dot" published in Nature Communications
 - (a) https://www.esa.int/Science_Exploration/Space_Science/Webb/Webb_spots_greedy_supermassive_black_hole_in_early_Universe
2. Firefly Sparkle - star clusters in a lensed Milky Way progenitor at $z = 8.3$
 - (a) <https://science.nasa.gov/missions/webb/found-first-actively-forming-galaxy-as-lightweight-as-young-milky-way/>
3. Detailed spatially resolved analysis of gravitationally lensed "Question Mark Galaxy"
 - (a) <https://science.nasa.gov/missions/webb/nasas-webb-reveals-distorted-galaxy-forming-cosmic-question-mark/>
4. Globular clusters in a lensed $z = 1.3$ "Sparkler Galaxy"
 - (a) <https://www.artsci.utoronto.ca/news/researchers-help-reveal-galaxy-sparkling-universe-oldest-star-clusters>
5. "Universe Breakers" don't live up to the name
 - (a) <https://news.smu.ca/news/2024/4/22/new-data-challenge-early-jwst-claims-about-the-age-of-the-universe>
6. $z = 5$ merging galaxies show how high- z galaxies grow
 - (a) <https://news.smu.ca/news/2023/9/21/researchers-use-the-james-webb-space-telescope-to-view-growing-baby-galaxy>
7. Redshift Catalog from first JWST data
 - (a) <https://news.smu.ca/news/2023/10/31/smu-researcher-leads-team-in-discovering-galaxies-with-james-webb-space-telescope?rq=noirot>

ADDITIONAL INFORMATION

Languages: English (Native), Slovenian (A1)

PUBLICATIONS (55 TOTAL, 5 AS FIRST AUTHOR; ADS)

As of January 2026, these works have 1743 citations with an h-index of 25. In the list below, my name is bolded.

First author

5. **Martis, N. S.** Withers, S. Bradač, M., et al., "CANUCS/Technicolor: JWST Medium-band Photometry Finds Half of the Star Formation at $z > 7.5$ Is Obscured", 2025, 990 p. 83.
4. **Martis, N. S.** Sarrouh, G. T. E. Willott, C. J., et al., "Modeling and Subtracting Diffuse Cluster Light in JWST Images: A Relation between the Spatial Distribution of Globular Clusters, Dwarf Galaxies, and Intracluster Light in the Lensing Cluster SMACS 0723", 2024, 975 p. 76.
3. **Martis, N. S.** Marchesini, D. M. Muzzin, A., et al., "Identifying and characterizing the most heavily dust-obscured galaxies at $1 \leq z \leq 4$ ", 2023, 518 pp. 4961–4975.
2. **Martis, N. S.** Marchesini, D. M. Muzzin, A., et al., "Stellar and Dust Properties of a Complete Sample of Massive Dusty Galaxies at $1 \leq z \leq 4$ from MAGPHYS Modeling of UltraVISTA DR3 and Herschel Photometry", 2019, 882 p. 65.
1. **Martis, N. S.** Marchesini, D. Brammer, G. B., et al., "The Evolution of the Fractions of Quiescent and Star-forming Galaxies as a Function of Stellar Mass Since $z = 3$: Increasing Importance of Massive, Dusty Star-forming Galaxies in the Early Universe", 2016, 827 p. L25.

Second or third author

4. Sarrouh, G. T. E. Asada, Y. **Martis, N. S.**, et al., “CANUCS/Technicolor Data Release 1: Imaging, Photometry, Slit Spectroscopy, and Stellar Population Parameters”, 2026, 282 p. 3.
3. Tripodi, R. **Martis, N.** Markov, V., et al., “Extreme properties of a compact and massive accreting black hole host in the first 500 Myr”, 2025, *Nature Communications* 16 p. 9830.
2. Desprez, G. **Martis, N. S.** Asada, Y., et al., “ Λ CDM not dead yet: massive high- z Balmer break galaxies are less common than previously reported”, 2024, 530 pp. 2935–2952.
1. Harshan, A. Tripodi, R. **Martis, N. S.**, et al., “Detailed Study of Stars and Gas in a $z = 8.3$ Massive Merger with Extreme Dust Conditions”, 2024, 977 p. L36.

Contributing author

46. Asada, Y. Willott, C. J. Muzzin, A., et al., (**including N. S. M.**) “Earliest Galaxy Evolution in the CANUCS+Technicolor Fields: Galaxy Properties at $z \sim 1016$ Seen with the Full NIRCам Medium- and Broadband Filters”, 2026, 996 p. 115.
45. Coulter, D. A. Larison, C. Pierel, J. D. R., et al., (**including N. S. M.**) “A spectroscopically confirmed, strongly lensed, metal-poor Type II supernova at $z = 5.13$ ”, 2026, *arXiv e-prints* arXiv:2601.04156.
44. Yanagisawa, H. Ouchi, M. Golubchik, M., et al., (**including N. S. M.**) “VENUS: Two Faint Little Red Dots Separated by ~ 70 pc Hidden in a Single Lensed Galaxy at $z \sim 7$ ”, 2026, *arXiv e-prints* arXiv:2601.06015.
43. Antwi-Danso, J. Muzzin, A. Robbins, L., et al., (**including N. S. M.**) “An Ancient Descendant of the First Galaxies”, 2025, *arXiv e-prints* arXiv:2512.03154.
42. Asada, Y. Desprez, G. Willott, C. J., et al., (**including N. S. M.**) “Improving Photometric Redshifts of Epoch of Reionization Galaxies: A New Empirical Transmission Curve with Neutral Hydrogen Damping Wing Ly α Absorption”, 2025, 983 p. L2.
41. Bradač, M. Judež, J. Willott, C., et al., (**including N. S. M.**) “Star Formation under a Cosmic Microscope: Highly Magnified $z = 11$ Galaxy behind the Bullet Cluster”, 2025, 995 p. L74.
40. Estrada-Carpenter, V. Sawicki, M. Abraham, R., et al., (**including N. S. M.**) “Metal-poor Star-forming Clumps in Cosmic Noon Galaxies: Evidence for Gas Inflow and Chemical Dilution Using JWST NIRISS”, 2025, 991 p. 188.
39. Harris, W. E. Reina-Campos, M. Keatley, K. E., et al., (**including N. S. M.**) “JWST Photometry of Globular Cluster Populations in MACS0417.5–1154”, 2025, 993 p. 210.
38. Markov, V. Bradač, M. Estrada-Carpenter, V., et al., (**including N. S. M.**) “Resolving dust and Ly α emission in a lensed galaxy at the epoch of reionization with JWST/CANUCS”, 2025, *arXiv e-prints* arXiv:2512.13778.
37. Markov, V. Gallerani, S. Pallottini, A., et al., (**including N. S. M.**) “Unveiling the trends between dust attenuation and galaxy properties at $z \sim 2–12$ with the James Webb Space Telescope”, 2025, 702 A33.
36. Mérida, R. M. Gaspar, G. Asada, Y., et al., (**including N. S. M.**) “The rise and fall of Little Red Dots could be driven by the environment”, 2025, *arXiv e-prints* arXiv:2510.06408.
35. Mérida, R. M. Gaspar, G. Sawicki, M., et al., (**including N. S. M.**) “Possible environmental quenching in an interacting little red dot pair at $z \sim 7$ ”, 2025, 698 A317.
34. Mérida, R. M. Sawicki, M. Iyer, K. G., et al., (**including N. S. M.**) “Probing the Star Formation Main Sequence down to $10^7 M_\odot$ at $1 < z < 9$ ”, 2025, *arXiv e-prints* arXiv:2509.22871.
33. Muzzin, A. Suess, K. A. Marchesini, D., et al., (**including N. S. M.**) “MINERVA: A NIRCам Medium Band and MIRI Imaging Survey to Unlock the Hidden Gems of the Distant Universe”, 2025, *arXiv e-prints* arXiv:2507.19706.
32. Rihtaršič, G. Bradač, M. Desprez, G., et al., (**including N. S. M.**) “CANUCS: Constraining the MACS J0416.1–2403 strong lensing model with JWST NIRISS, NIRSpec, and NIRCам”, 2025, 696 A15.

31. Sok, V. Muzzin, A. Tan, V. Y. Y., et al., **(including N. S. M.)** “The Stellar Mass and Age Distributions of Star-Forming Clumps at $0.5 < z < 5$ in JWST CANUCS: Implications for Clump Formation and Destruction”, 2025, *arXiv e-prints* arXiv:2509.25363.
30. Tan, V. Y. Y. Muzzin, A. Sarrouh, G. T. E., et al., **(including N. S. M.)** “Resolved Mass Assembly and Star Formation in Milky Way Progenitors since $z = 5$ from JWST/CANUCS: From Clumps and Mergers to Well-ordered Disks”, 2025, 994 p. 94.
29. Tripodi, R. Bradač, M. D’Eugenio, F., et al., **(including N. S. M.)** “A Deep Dive down the Broad-line Region: Permitted O I, Ca II, and Fe II Emission in an Active Galactic Nucleus Little Red Dot at $z = 5.3$ ”, 2025, 994 p. L6.
28. Willott, C. J. Asada, Y. Iyer, K. G., et al., **(including N. S. M.)** “In Search of the First Stars: An Ultra-compact and Very-low-metallicity Ly α Emitter Deep within the Epoch of Reionization”, 2025, 988 p. 26.
27. Zhang, Z. Li, M. Oguri, M., et al., **(including N. S. M.)** “Little red dot variability over a century reveals black hole envelope via a giant Einstein cross”, 2025, *arXiv e-prints* arXiv:2512.05180.
26. Asada, Y. Sawicki, M. Abraham, R., et al., **(including N. S. M.)** “Bursty star formation and galaxy-galaxy interactions in low-mass galaxies 1 Gyr after the Big Bang”, 2024, 527 pp. 11372–11392.
25. Bradač, M. Strait, V. Mowla, L., et al., **(including N. S. M.)** “Star Formation at the Epoch of Reionization with CANUCS: The Ages of Stellar Populations in MACS1149-JD1”, 2024, 961 p. L21.
24. Estrada-Carpenter, V. Sawicki, M. Brammer, G., et al., **(including N. S. M.)** “When, where, and how star formation happens in a galaxy pair at cosmic noon using CANUCS JWST/NIRISS grism spectroscopy”, 2024, 532 pp. 577–591.
23. Gledhill, R. Strait, V. Desprez, G., et al., **(including N. S. M.)** “CANUCS: An Updated Mass and Magnification Model of A370 with JWST”, 2024, 973 p. 77.
22. Harshan, A. Bradač, M. Abraham, R., et al., **(including N. S. M.)** “CANUCS: UV and ionizing properties of dwarf star-forming galaxies at $z = 5-7$ ”, 2024, 532 pp. 1112–1125.
21. Mowla, L. Iyer, K. Asada, Y., et al., **(including N. S. M.)** “Formation of a low-mass galaxy from star clusters in a 600-million-year-old Universe”, 2024, 636 pp. 332–336.
20. Nedkova, K. V. Häußler, B. Marchesini, D., et al., **(including N. S. M.)** “Bulge+disc decomposition of HFF and CANDELS galaxies: UVJ diagrams and stellar mass-size relations of galaxy components at $0.2 \leq z \leq 1.5$ ”, 2024, 532 pp. 3747–3777.
19. Sarrouh, G. T. E. Muzzin, A. Iyer, K. G., et al., **(including N. S. M.)** “Exposing Line Emission: The Systematic Differences of Measuring Galaxy Stellar Masses with JWST NIRCам Medium versus Wide Band Photometry”, 2024, 967 p. L17.
18. Willott, C. J. Desprez, G. Asada, Y., et al., **(including N. S. M.)** “A Steep Decline in the Galaxy Space Density beyond Redshift 9 in the CANUCS UV Luminosity Function”, 2024, 966 p. 74.
17. Antwi-Danso, J. Papovich, C. Leja, J., et al., **(including N. S. M.)** “Beyond UVJ: Color Selection of Galaxies in the JWST Era”, 2023, 943 p. 166.
16. Asada, Y. Sawicki, M. Desprez, G., et al., **(including N. S. M.)** “JWST catches the assembly of a $z = 5$ ultra-low-mass galaxy”, 2023, 523 pp. L40–L45.
15. Matharu, J. Muzzin, A. Sarrouh, G. T. E., et al., **(including N. S. M.)** “A First Look at Spatially Resolved Balmer Decrements at $1.0 < z < 2.4$ from JWST NIRISS Slitless Spectroscopy”, 2023, 949 p. L11.
14. Noirot, G. Desprez, G. Asada, Y., et al., **(including N. S. M.)** “The first large catalogue of spectroscopic redshifts in Webb’s first deep field, SMACS J0723.3-7327”, 2023, 525 pp. 1867–1884.
13. Strait, V. Brammer, G. Muzzin, A., et al., **(including N. S. M.)** “An Extremely Compact, Low-mass Galaxy on its Way to Quiescence at $z = 5.2$ ”, 2023, 949 p. L23.
12. Vanzella, E. Loiacono, F. Bergamini, P., et al., **(including N. S. M.)** “An extremely metal-poor star complex in the reionization era: Approaching Population III stars with JWST”, 2023, 678 A173.

11. Withers, S. Muzzin, A. Ravindranath, S., et al., **(including N. S. M.)** “Spectroscopy from Photometry: A Population of Extreme Emission Line Galaxies at $1.7 < z < 6.7$ Selected with JWST Medium Band Filters”, 2023, 958 p. L14.
10. Marsan, Z. C. Muzzin, A. Marchesini, D., et al., **(including N. S. M.)** “The Number Densities and Stellar Populations of Massive Galaxies at $3 < z < 6$: A Diverse, Rapidly Forming Population in the Early Universe”, 2022, 924 p. 25.
9. Mowla, L. Iyer, K. G. Desprez, G., et al., **(including N. S. M.)** “The Sparkler: Evolved High-redshift Globular Cluster Candidates Captured by JWST”, 2022, 937 p. L35.
8. Tan, V. Y. Y. Muzzin, A. Marsan, Z. C., et al., **(including N. S. M.)** “Resolved Stellar Mass Maps of Galaxies in the Hubble Frontier Fields: Evidence for Mass Dependency in Environmental Quenching”, 2022, 933 p. 30.
7. Willott, C. J. Doyon, R. Albert, L., et al., **(including N. S. M.)** “The Near-infrared Imager and Slitless Spectrograph for the James Webb Space Telescope. II. Wide Field Slitless Spectroscopy”, 2022, 134 p. 025002.
6. Nedkova, K. V. Häußler, B. Marchesini, D., et al., **(including N. S. M.)** “Extending the evolution of the stellar mass-size relation at $z \leq 2$ to low stellar mass galaxies from HFF and CANDELS”, 2021, 506 pp. 928–956.
5. Silva, A. Marchesini, D. Silverman, J. D., et al., **(including N. S. M.)** “Galaxy Mergers up to $z < 2.5$. II. AGN Incidence in Merging Galaxies at Separations of 3-15 kpc”, 2021, 909 p. 124.
4. Roebuck, E. Sajina, A. Hayward, C. C., et al., **(including N. S. M.)** “Simulations Find Our Accounting of Dust-obscured Star Formation May Be Incomplete”, 2019, 881 p. 18.
3. Shipley, H. V. Lange-Vagle, D. Marchesini, D., et al., **(including N. S. M.)** “HFF-DeepSpace Photometric Catalogs of the 12 Hubble Frontier Fields, Clusters, and Parallels: Photometry, Photometric Redshifts, and Stellar Masses”, 2018, 235 p. 14.
2. Silva, A. Marchesini, D. Silverman, J. D., et al., **(including N. S. M.)** “Galaxy Mergers up to $Z < 2.5$. I. The Star Formation Properties of Merging Galaxies at Separations of 3-15 kpc”, 2018, 868 p. 46.
1. Brammer, G. B. Marchesini, D. Labbé, I., et al., **(including N. S. M.)** “Ultra-deep K s -band Imaging of the Hubble Frontier Fields”, 2016, 226 p. 6.

REFERENCES

1. **Danilo Marchesini, danilo.marchesini@tufts.edu**

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Relationship: Thesis Advisor

2. **Maruša Bradač, Marusa.Bradac@fmf.uni-lj.si**

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Relationship: Collaborator